

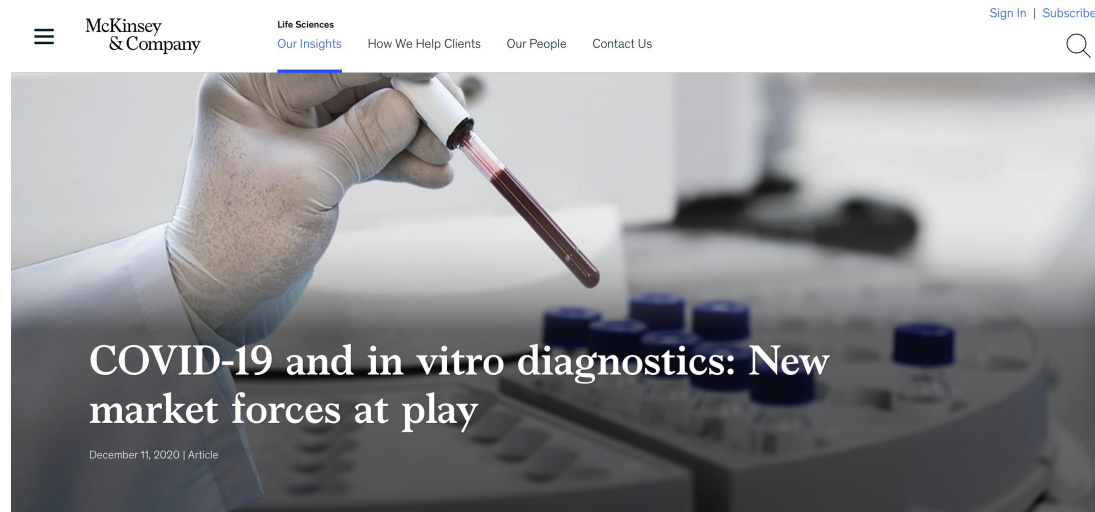
Valuation and Analysis of DiaSorin Group

Filippo Boggetti



In Vitro Diagnostic (IVD) Market Highlights

- WW In Vitro Diagnostics value 60 billions euro
- > 70% of clinical decision are based on IVD data
- After COVID-19 pandemic white economy is considered strategic for the developed countries
- EU has started a program to reshore production and protect the existing industries through golden share options



The high demand for in vitro diagnostics could seed structural shifts that will have long-term implications for diagnostic-test manufacturers.

STUDY
Requested by the INTA committee



Post Covid-19 value chains:
options for reshoring production
back to Europe
in a globalised economy



Policy Department for External Relations
Directorate General for External Policies of the Union
PE 653.626 – March 2021

EN

DiaSorin Overview



Innovative company Italy based operating in the field of In Vitro Diagnostics, within which **develops, produces and markets tests** for diagnosing changes to the patient's health, following infection, hormonal changes or cancer.

- Two market segments: **Molecular and Immunodiagnostic**
- Recent acquisition (April 2021) of **Luminex**, molecular firm specialized in Assays and multiplexing

Financial summary
(Diasorin 2020)

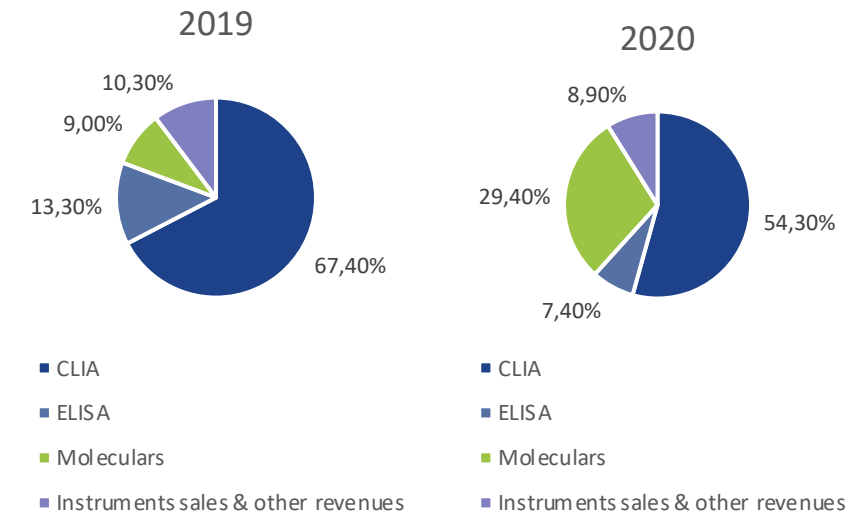
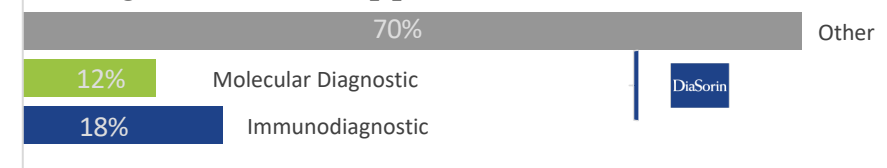
Revenues 2020	858.9 m	Revenues 2020	340.15 m
EBITDA adj	383.2 m	EBITDA	59.430 m
EBIT	326.305 m	EBIT	34.588 m

Financial summary
(Luminex 2020)

Revenues 2020	340.15 m
EBITDA	59.430 m
EBIT	34.588 m

Diagnostic Market

Diagnostic value approx: 60 billion euro



Data on Revenues segmentation does not include Luminex and 2020 revenues show the impact on molecular revenues of SARS-COV-2

Market Considerations

Production:

- Immunodiagnostic: end product and, in most cases, all required components
- Molecular: buy reaction buffers, enzymes, oligonucleotides
- High regulatory requirements (new IVDR) and strict product post market surveillance requires zero defects manufacturing

Distribution:

- Opening tenders, private customers, letters of offer. Highly regulated sector in which less contractual power is given to suppliers
- DiaSorin uses **distribution contracts** where no subsidiary is present

Business model:

- Analytical instruments are usually loaned (at really low rates) based on a reagents supply contract. After sales support and HW servicing is mostly free. (**Razor-razorblade business model**)

Covid Testing (*):

- LIAISON(IgG, IgM), SIMPLEXA (molecular), Antigen(LIAISON, SARS – CoV2 – AG)

(*) covid tests impacted DiaSorin revenues in 2020 and so this revenues should be carefully valued.

Products I

Immunodiagnosics

Technology that allows to detect and quantify the presence of specific molecules in biological fluids. Fluids can be either loaded on **Clia** platforms (Chemiluminescence) or **Elisa platforms** (Colorimetric). Difference between the two is the type of detection (**linearity and sensitivity of the test**). **Immunodiagnostic** business represent **44%** of revenues after the acquisition.

Clia System: *Closed system*

Automated technology which is the one mostly used in laboratories. Products in this line are created in order to target different laboratories, from smaller ones to bigger ones with the need of integration and continuous loading of samples, reagents and consumables (*Walk away*).



LIAISON



LIAISON 



LIAISON 



LIAISON 
LAS

Elisa System: *Open platform*

Clia based on 96 well plates is a semi automated technology that requires a manual process to prepare the sample before it is loaded on the analyser.



Freedom EVOlyzer® 2-150/8



ETI-MAX3000

Products II

Molecular Diagnostic

Technology that allows to diagnose a pathology by detecting specific RNA and DNA sequences (nucleic acids) in patients' biological fluids or abnormal cells. The line of products offers thermocyclers for amplification of DNA and RNA through quantitative PCR.

Liaison MDX



Thermocycler with two consumable disc options: the 8-well Direct Amplification Disc for sample-to-answer testing and the 96-well Universal Disc for higher volume testing. Run real-time PCR for qualitative, quantitative and multi-analyte detection. The instrument can perform both IVD as well as Laboratory Developed Tests.



Liaison IAM:



Instrument developed by DiaSorin specifically to support the isothermal Q-LAMP technology for amplification of Nucleic Acids.



Vergine, Xmap, Flexmap, Luminex 200, Magpix



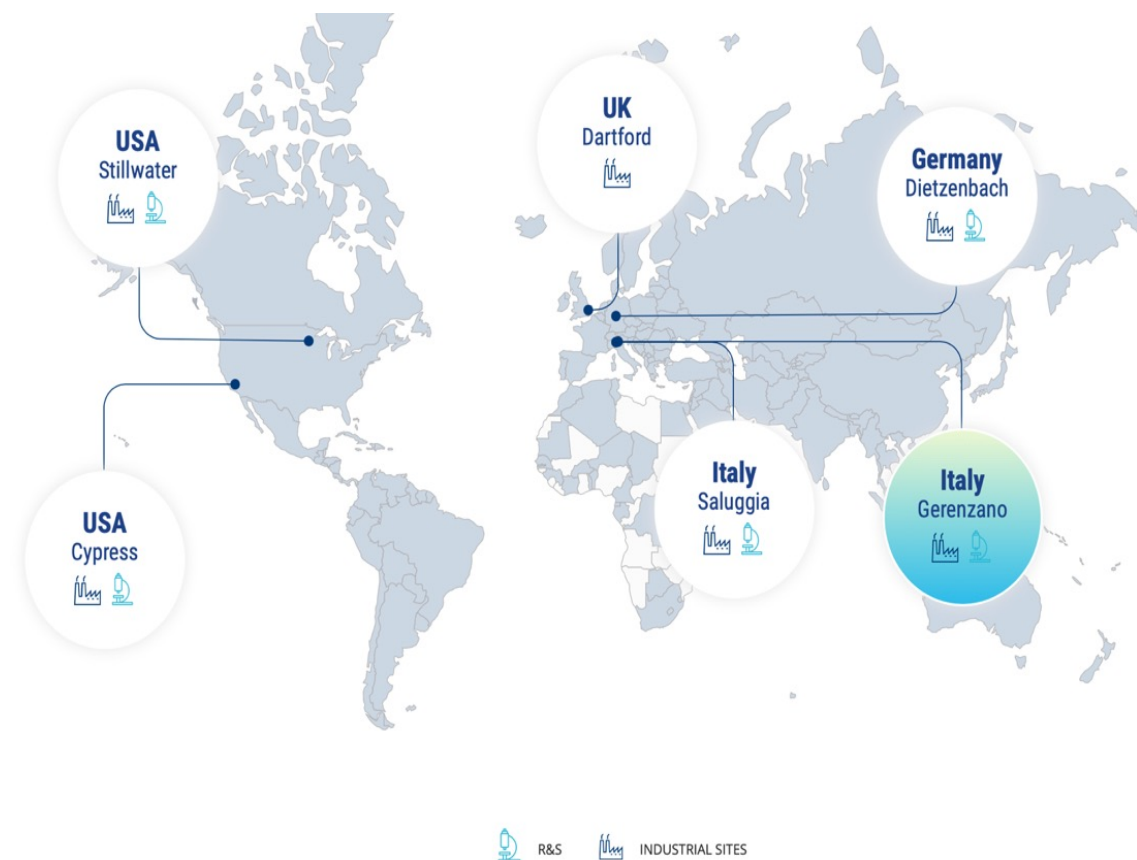
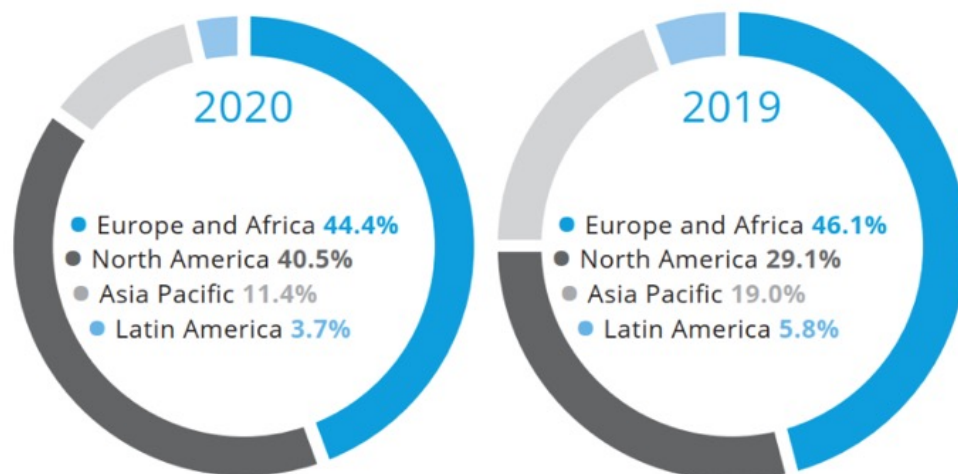
Luminex offers a great range of molecular testing based on Xmap or Xtag technologies that complete DiaSorin offer



Geographical Presence DiaSorin

Breakdown of revenues by geographic region

(€ thousands)	2020	2019	% Change at current exchange rates	% Change at constant exchange rates
Europe and Africa	391,430	325,289	20.3%	20.5%
USA and Canada	357,117	205,792	73.5%	77.0%
Asia Pacific	100,598	134,267	-25.1%	-23.7%
Latin America	32,160	40,970	-21.5%	-7.5%
Total on a like-for-like basis	881,305	706,319	24.8%	27.0%

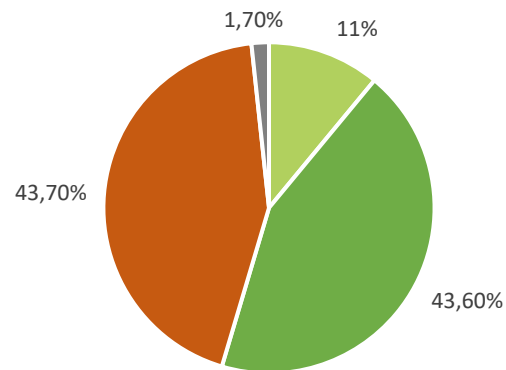


Luminex Acquisition - Overview

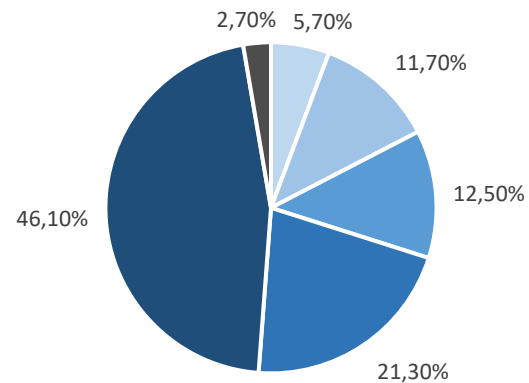
- Biotech company providing molecular, proteomic, and cellular analysis tools and tests to life science and diagnostics market.
- Leading global provider for Molecular Diagnostics, specialized in **multiplexing technology**
- Applications throughout Life Science industries, including pharmaceuticals, academia and research
- **Proprietary testing technologies**, supporting broad biological test menu
- Extensive **platform placements** (~25,000 units)
- **850+ patents owned** on differentiated platforms and assays
- **900+** Molecular diagnostics **active clients**

Revenues 2020 USD 417 m

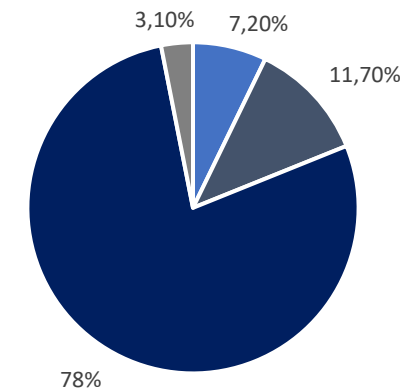
By technology



By product



By geography



■ MDX Single/Low Plex ■ MDX multiplexing ■ Life Since ■ Other ■ Service ■ Consumables ■ Royalties ■ Systems ■ Assays ■ Other

■ Asia ■ Europe ■ USA ■ Other

Luminex Acquisition - Rationale



Strategic transaction identified as part of a review on **multiplex** opportunities and analysis of existing and emerging technology offerings

Opportunity in line with DiaSorin strategic priorities, providing:



- access to proven **molecular multiplexing** technology to be used in unique testing panels
- critical mass in the molecular diagnostics space
- **broader presence** in the U.S.
- access to **new partnerships** and business development opportunities through life science offerings
- **broader access** to DiaSorin **Clia** offerings, especially in the U.S. hospital setting

Opportunity in line with DiaSorin strategic priorities, providing:



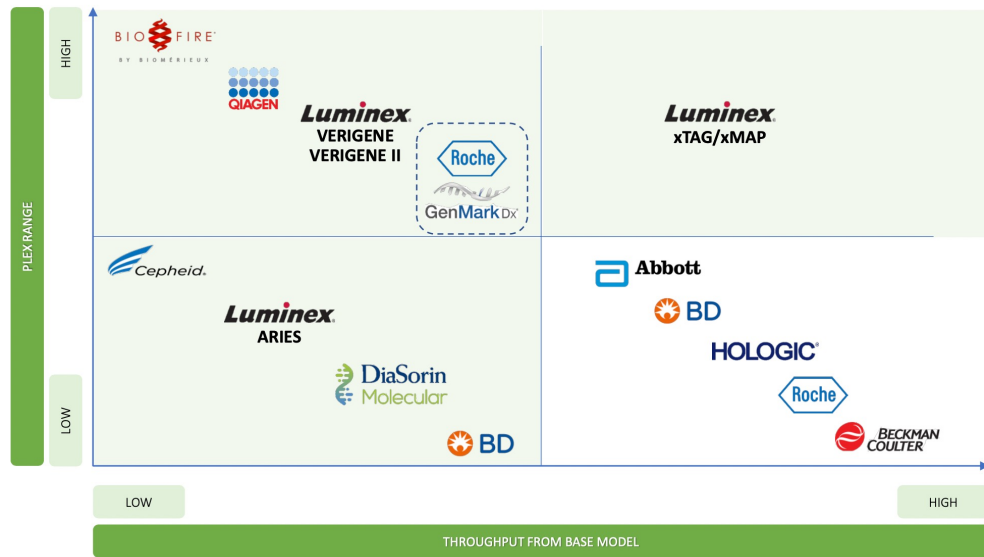
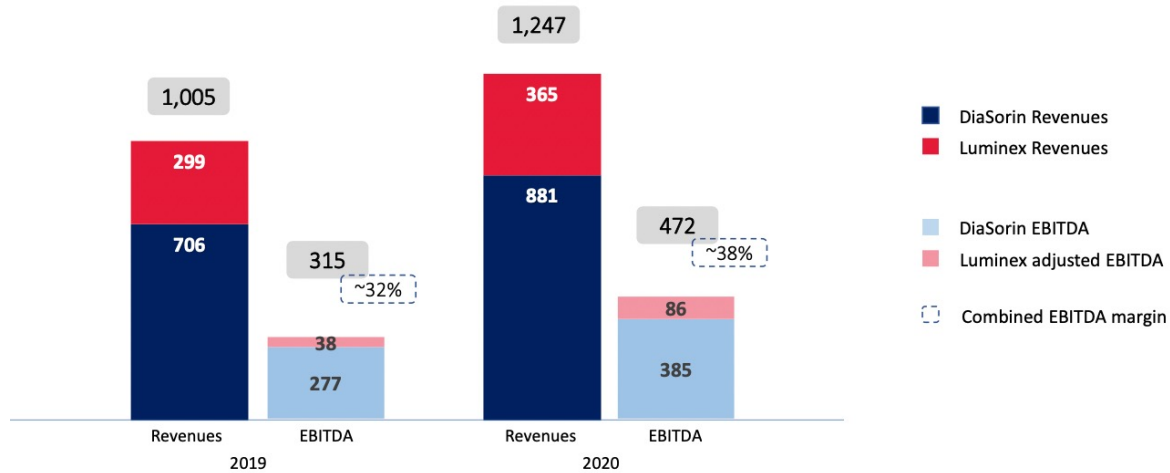
- net purchase price: USD 37.00 per share, equal to an equity value of approximately USD 1.8 bn and an enterprise value of approximately USD 1.8 bn
- term loan for USD 1.1 m
- bridge loan for USD 500 k
- obligations for the remaining part

Creating shareholder **value** through:

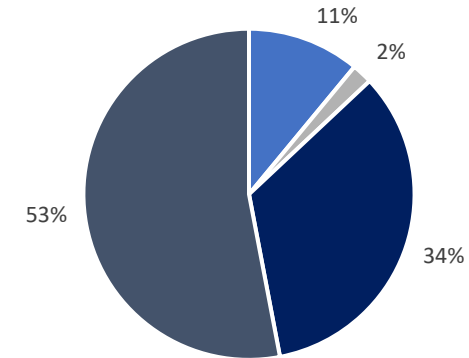


- **EPS** immediately accretive post closing
- attractive return on invested capital profile
- estimated **cost synergies** of approximately **USD 55 m** within 3 years after closing (*) [used to perform valuation]

Overview on New Positioning

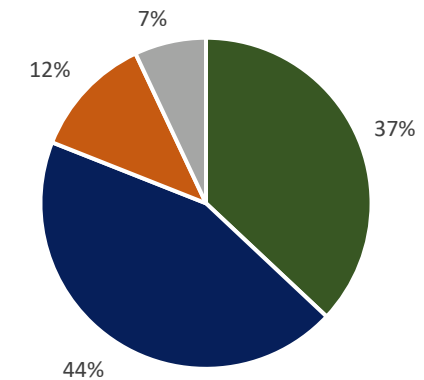


By geography



■ Asia Pacific ■ Latin America ■ Europe & Africa ■ USA & Canada

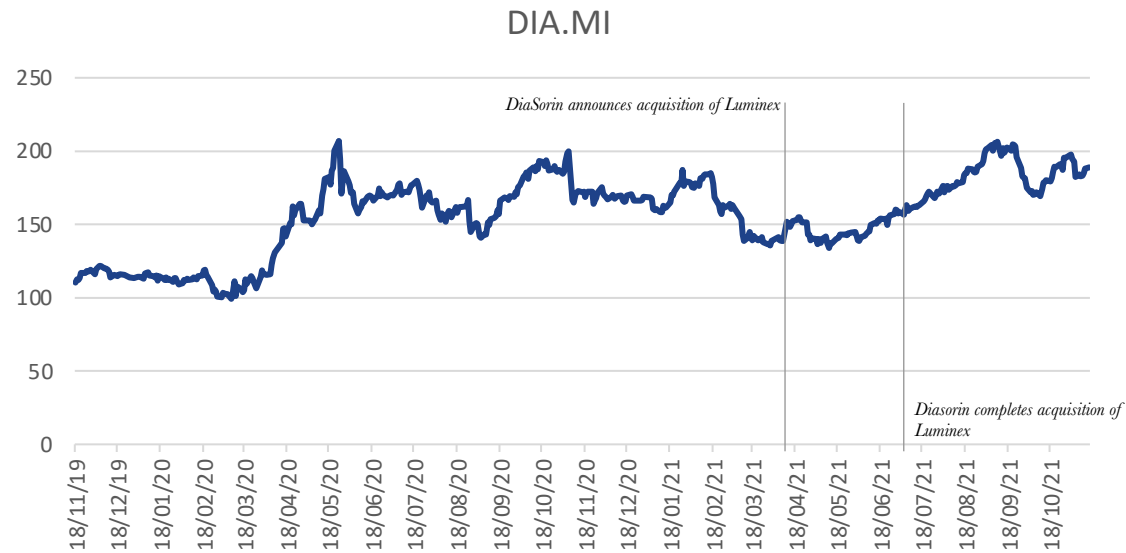
By technology



■ MDX ■ Immuno ■ Life science ■ Other



Market Data



€191.7

LAST PRICE

1.46

% CHANGE

6.24

3 MONTH PERFORMANCE

8.72

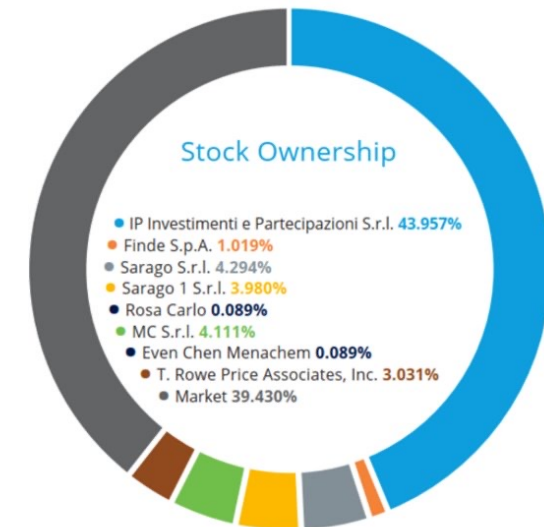
1 YEAR PERFORMANCE

Data is referring to 16th November 2021

Capitalization 10.66 Mrd

P/E 35.62

D/P 0.52%



The firm still remains proprietary based since Investimenti Piemonte “*IP Investimenti e partecipazioni*” is a private fund held by the founder of DiaSorin **Gustavo De Negri** who is also president of the company. **Menachem and Carlo Rosa** (CEO) led the MBO in 2000.

Playing with Mirrors

As to calculate the WACC, we first tried to obtain the beta of the firm by our own through the **Capital Asset Pricing Model**.

Unfortunately, we encountered an unsatisfactory result: being a healthcare firm COVID related, DiaSorin has presented a **negative correlation to the market** in the recent past.

In the regression we have performed through MATLAB we have used as the endogenous variable the monthly stock returns of the last two years of DiaSorin and the returns of the italian index FTSE IT ALL-SHARE INDEX as the regressor.

The output of the regression is the following:

- *Equity market risk premium* = 8.60%
- α = 30.52%
- β = -0.12

Given this estimates, we would have obtained a WACC extremely low (below 0.5%) that would not have described well the reality.

```
% Importing data

[prices, tickers] = xlsread("ITLMS DIA.xlsx")

% From prices to returns

nobs = size(prices, 1)
returns = prices(2:nobs,:) ./ prices(1:(nobs-1),:) - 1
n = 12

% return of a risk-free asset: rf

rf = 0.01

% Set up of the CAPM (with intercept)

emarket = returns(:, 1) * n - rf
X = [ones(length(emarket), 1), emarket]

Y = returns(:, 2) * n

[B, BINT, R, RINT, STATS] = regress(Y, X, 0.05)
```


Weighted Average Cost of Capital

The estimates of the expected inflations are the **current benchmark of the FED and the ECB** for next year.

The US risk free rate is the yield of the 10Y T-Bill of early November.

The equity market risk premium (for Italy) and the leveraged equity beta (for the Healthcare Products sector) were provided by the NYU finance professor Aswath Damodaran.

Cost of equity	7.09%
US risk free rate	1.50%
Expected US inflation	2.00%
Expected Eurozone inflation	1.70%
Risk free rate	1.21%
Equity market risk premium	6.23%
Leveraged equity beta	0.95

The cost of debt and the tax rate were provided by the pro-forma 2020's financial statement in which Luminex and DiaSorin were consolidated and **in which the cost of the acquisition was present.**

The value of equity is DiaSorin market capitalization at the end of the last fiscal year.

The weighted average cost of capital computed with this inputs is 6.6%

Cost of debt	2.91%
Net financial expense	31,128
Net financial liabilities	1,068,364

Tax rate	24.44%
Income tax	78,157
Earnings before tax	319,816

Enterprise value	
Equity	89.91%
Net debt	10.09%

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
import statsmodels.api as sm
import seaborn as sns

# Importing data

filename = "Modified Jones Model.csv"
data = pd.read_csv(filename)

# Scale the independent variables and the response variable

inv_asset = np.ones(data.shape[0]) / np.array(data["Asset t - 1"])
drev_less_drec = np.array(data["Delta Rev t - Delta Rec t"]) / np.array(data["Asset t - 1"])
ppe = np.array(data["PPE t"]) / np.array(data["Asset t - 1"])

accruals = np.array(data["Accruals t"]) / np.array(data["Asset t - 1"])

# Aggregate all the independent variables in a single matrix

features = np.transpose(np.matrix([inv_asset, drev_less_drec, ppe]))

# Create the linear model and analysis of the result

model = sm.OLS(accruals, features)
result = model.fit()

print(result.summary())

# Make predictions with the model

predictions = result.predict(features)

# Scatter the fitted values vs the actual values

plt.scatter(predictions, accruals, alpha = 0.2, color= 'b')

# Plot the perfect prediction line

xmin, xmax = plt.xlim()
plt.plot(np.arange(xmin, xmax, 0.01), np.arange(xmin, xmax, 0.01), c='k')

# Set the axis labels and show the plot

plt.xlabel('accruals predicted')
plt.ylabel('actual accruals')

plt.show()

# Outline the correlations between the different variables

dataframe = pd.DataFrame(features)
labels = ["One / Asset t-1", "(DR t - DR t) / Asset t-1", "PPE t / Asset t-1"]
dataframe.columns = labels
dataframe["Accruals / Asset t-1"] = accruals

corr = dataframe.corr()

# Plot heatmap of correlation matrix

sns.heatmap(corr, annot= True, annot_kws = {"size": 12.5})

# fix ticklabel directions and size

plt.yticks(rotation = 0, size = 12.5)
plt.xticks(rotation = 90, size = 12.5)

plt.tight_layout()

plt.show()
```

Modified Jones Model

The aim of the regression is to estimate the non-discretionary accruals in the firm's balance sheet, considering its revenues, accounts receivable and physical assets, as to verify if there has been earnings manipulation by the management.

$$\frac{TA_{i,t}}{A_{i,t-1}} = \beta_i \left[\frac{1}{A_{i,t-1}} \right] + \beta_{1i} \left[\frac{(\Delta REV_{it} - \Delta REC_{it})}{A_{i,t-1}} \right] + \beta_{2i} \left[\frac{PPE_{it}}{A_{i,t-1}} \right] + \varepsilon_{it}$$

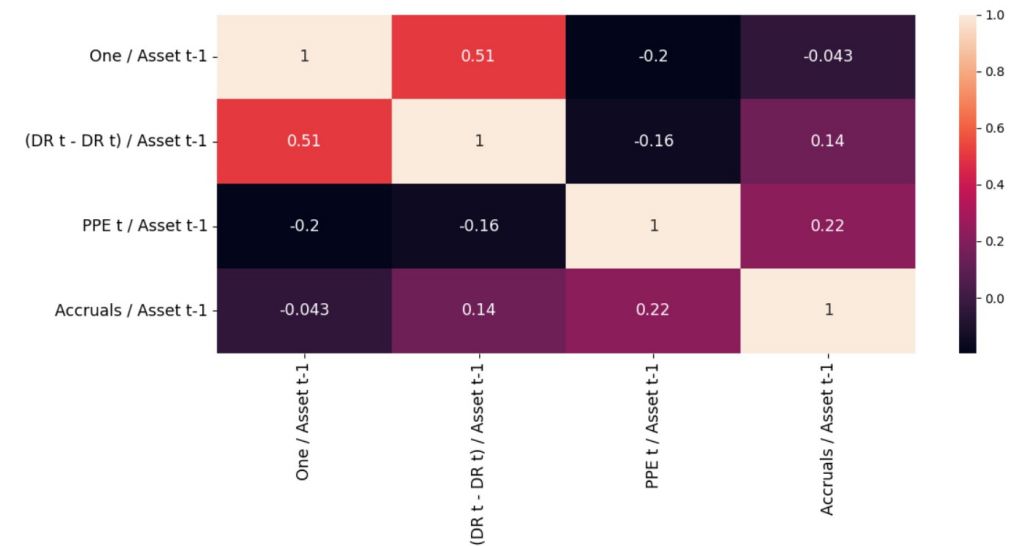
The differential between the actual accruals and the predicted accruals of a firm are considered as discretionary accruals, possibly signalling *earnings management*.

$$\frac{DA_{i,t}}{A_{i,t-1}} = \frac{TA_{i,t}}{A_{i,t-1}} - b_i \left[\frac{1}{A_{i,t-1}} \right] - b_{1i} \left[\frac{(\Delta REV_{it} - \Delta REC_{it})}{A_{i,t-1}} \right] - b_{2i} \left[\frac{PPE_{it}}{A_{i,t-1}} \right]$$

Econometric Analysis

We have applied the modified model on 40 firms of the Healthcare Products sector to test the earnings quality of DiaSorin

- The regression has a R^2_{adj} of 32.1% .
- $\left[\frac{1}{A_{i,t-1}} \right]$ and $\left[\frac{PPE_{it}}{A_{i,t-1}} \right]$ present a p -value lower than 5%.
- $\left[\frac{(\Delta REV_{it} - \Delta REC_{it})}{A_{i,t-1}} \right]$ is not statistically significant due to multicollinearity, as depicted by the heatmap of the correlations.



Through the regression we have estimated the discretionary accruals of DiaSorin, which represents in absolute value only 3.19% of the total assets of the previous fiscal year, against a sample's mean value of 4.84%.

Given this estimate, we can affirm the **earnings quality** of DiaSorin is high and there has not been any earnings creation with accounting methods (e.g., *bleedbacks*).

Market share acquisition on Immunodiagnostic	2%
Δp Clia	-0,4% in 2021 decreasing to -0,1% in 2030
Elisa growth	3% to stable values
Molecular (No covid) growth diasorin segment	4% Historical values
Instruments and sales Dia	Sales/(CLIA+Elisa), historical mean
System sales Luminex	anchored on assays 33,3% with linear decreasing to 31% (less molecular sales that have higher margins)
Ebit/Sales	23%
Tax rate dia	21,74% historical mean
Inventory/Sales	73 days mean no movement
average collection time Dia	31 days no movement
average payables time Dia	
average collection time Luminex	52 days mean no movement
average payables time Luminex	16 days no movement
Inventory/Sales	11,3% historical mean
Tax rate Luminex	32% no movement
CAGR molecular Luminex	12,35 mean reversion applied
Reveneues synergies on Luminex	2% managerial assumption
Royalties sales Luminex	Fixed
Depreciation for both Diasorin an luminex	1% year
Capex	Divided in Development and replacement anchoring on assets movement
	divided on steady and development, anchored to capex
Ammortization	

Assumptions

Assumptions are derived from market research as well as managerial assumption obtained by interviewing managers in the immunodiagnostic sector. From this analysis we forecasted a **reduction in Covid revenues from 2022** in the molecular sector.

To the right sales in the explicit forecast and on the right the WC scheme used, similar one has been used for Luminex. Assumption for CLIA, ELISA, Molecular MDX, Luminex revenues and their smaller segment reported in the upper scheme.

	Dati storici							Orizzonte esplicito				
	2015	2016	2017	2018	2019	2020	H 2021	2021	2022	2023	2024	2025
CLIA	371.391	396.611	423.291	440.516	476.059	478.549	286.571	549.262	602.088	660.658	725.653	797.841
ELISA	66.391	77.040	84.786	95.210	93.940	65.217	27.832	53.346	60.000	70.000	80.000	90.000
Moleculars	2.995	31.957	54.186	56.992	63.569	259.104	160.294	307.231	200.000	150.000	100.000	90.000
Instruments												
sale & other revenues	58.404	65.056	75.223	77.778	72.751	78.436	40.718	78.043	121.355	122.651	124.920	134.329
	15,7%	16,4%	17,8%	17,7%	15,3%	16,4%	14,2%	14,2%				
Totale	499.181	570.663	637.487	670.496	706.319	881.305	515.416	987.881	983.443	1.003.310	1.030.573	1.112.170

	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
System sales	28.177	35.496	32.228	34.922	62.557	57.806	65.335	74.709	66.322	75.838	85.201
System sales/ Assay revenue		30,31%	30,65%	24,95%	25,52%	53,23%	33,39%	33,01%	33,01%	33,01%	33,01%
Consumable sales	39.756	46.102	41.123	43.794	43.210	39.975	42.326	42.326	42.326	42.326	42.326
Royalty revenue	38.131	41.784	37.275	43.139	47.678	39.924	41.322	41.322	41.322	41.322	41.322
Assay revenue	92.970	115.799	129.165	136.868	117.525	173.100	197.935	226.334	200.926	229.753	258.122
Service revenue	8.773	10.261	9.564	10.619	19.951	19.067	20.145	23.036	20.450	23.384	26.271
Service revenue/ System sales	31,14%	28,91%	29,68%	30,41%	31,89%	32,98%	30,83%	30,83%	30,83%	30,83%	30,83%
Other revenue	10.536	7.307	6.270	6.481	6.958	10.276	7.971	7.971	7.971	7.971	7.971
TOTAL	218.341	256.749	255.625	275.823	297.880	340.149	375.035	415.698	379.317	420.594	461.214

WC Analisi

Dati storici		Orizzonte esplicito												
		2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Inventario (Ks)	Raw Materials	25.663	30.202	34.715	40.169	43.355	48.469	52.437	70.405					
	Write down provisions (RM)	(2.103)	(1.804)	(2.174)	(2.860)	(3.023)	(2.436)	(2.672)	(2.919)					
	Net Amount	23.560	28.398	32.541	37.309	40.332	46.033	49.765	67.486					
	Work in Progress	37.081	40.834	40.798	44.816	47.468	49.542	52.274	55.332					
	Write downs (WP)	(2.296)	(2.738)	(3.134)	(2.536)	(2.133)	(2.155)	(2.370)	(2.860)					
	Net Amount	34.785	38.076	37.664	42.280	45.335	47.387	49.904	52.472					
	Finished goods	29.666	36.426	37.250	51.755	58.982	68.994	73.863	75.906					
	Write down provisions (FG)	(1.572)	(1.580)	(1.262)	(2.474)	(1.947)	(1.958)	(2.405)	(4.630)					
	Net Amount	28.094	34.846	35.988	49.281	57.135	66.976	71.458	71.276					
	Totale Gross	92.410	107.462	112.763	136.740	149.805	166.945	178.574	201.643					
Totale inventario/sales	86.439	101.328	106.193	128.870	142.802	160.396	171.127	191.234	214.737	211.872	215.256	221.105	238.611	
				21,27%	22,58%	22,40%	25,92%	24,23%	21,70%	196.835	195.951	199.909	205.342	221.600
Trade receivables (Ks)	Trade receivables (*)	117.442	109.521	105.609	120.261	128.822	131.092	132.513	165.678	196.835	195.951	199.909	205.342	221.600
	Tempo medio d'incasso (NO IVA)	8.100	8.882	9.821	10.615	9.615	8.882	8.620	8.737	73	73	73	73	73
Other Current assets(Ks)	Other current assets	8.689	10.291	12.173	15.784	36.303	26.847	24.981	16.998	19.054	18.968	19.351	19.877	21.451
	Other su sales			2,44%	2,77%	5,69%	4,00%	3,54%	1,93%	1,9%	1,9%	1,9%	1,9%	1,9%
Other current financial assets (Ks)	Other current financial assets	34	24.963	58.179	0	13.974	23.422	44.588	126	141	141	143	147	159
Trade payables (Ks)	Trade payables	36.601	39.311	40.775	47.674	62.331	57.286	55.733	65.485	82.595	82.224	83.885	86.165	92.987
	Tempi medi d'incasso			30	30	30	31	29	27	31	31	31	31	31
Other current Liabilities (Ks) (**)	Other current Liabilities	26.303	30.573	32.837	41.870	49.689	50.520	51.134	60.688					
(**) in nota integrativa, non compatibile con bilancio poiché il bilancio AIDA non separa fra Other payables, Current tax Liabilities, Current Financial Liabilities cosa che in NOTA INTEGRATIVA viene fatto. Rislutano quindi maggiori e indicati sotto:														
Assets held for sale		4.000												
Other current Liabilities (Ks) (**)	Other current Liabilities	34.280	39.540	39.221	57.697	54.901	60.587	66.738	94.728	91.895	91.482	93.330	95.866	103.456
	Tempi medi			7,9%	10,1%	8,6%	9,0%	9,4%	10,7%	9,30%	9,30%	9,30%	9,30%	9,30%
Debiti	Debiti	6.752	3.266	2.444	26.512	23.888	20.579	8	0	0	0	0	0	0
				0,49%	4,65%	3,75%	3,07%	0,00%	0,00%					
WC		134.937	139.015	141.535	133.032	184.781	203.305	250.730	213.823	256.277	253.226	257.444	264.440	285.377
		10.533	(3.268)		(2.895)	13.704	15.504	7.152	51.569	42.454	(3.051)	4.219	6.996	20.937
AWC		-	4.078	2.520	(8.503)	51.749	18.524	47.425	(36.907)					
	Coerente	Coerente	Coerente	Coerente	Coerente	Coerente	Coerente	Coerente	Coerente					
		134.937	139.015	141.535	133.032	184.781	203.305	250.730	213.823	256.277	253.226	257.444	264.440	285.377

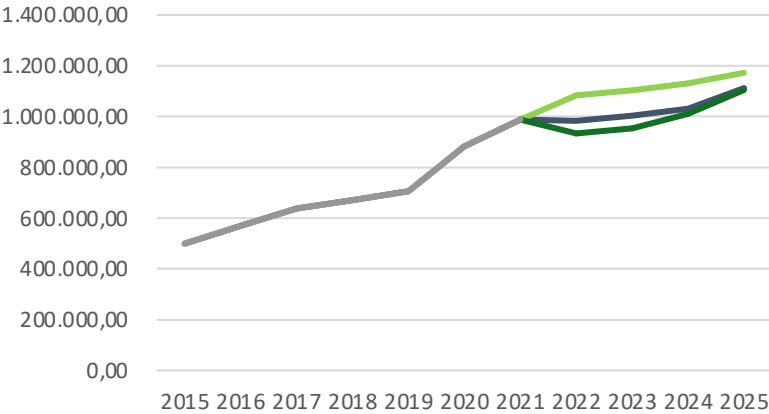
DCF Valuation I

The methodology used to forecast sales during the explicit forecast 2021-2025 is:

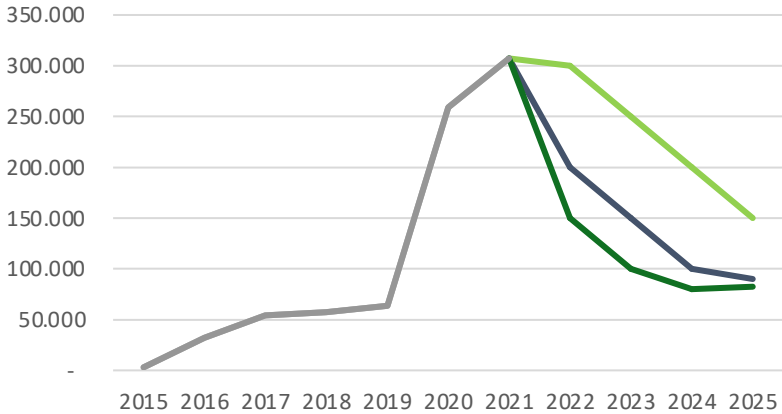
$$Rev_{t+1} = Rev_t(1 + \Delta q_{Firm})(1 + \Delta p_{Firm})$$
$$(1 + \Delta q_{Firm}) = (1 + \Delta demand_{Sector})(1 + \Delta market\ share)$$

Δp_{Firm} is estimated from **Goetzpartners market research**.
 $\Delta market\ share$ is estimated by **managerial assumptions** obtained by interviewing sector expert as well as probable **covid related revenues** developments. Over the explicit forecast we smooth the result till the long time growth. **Results** obtained over this forecast method **are also coherent** with an **historical growth forecasting** corrected for covid. Synergies have been valued as indicated before (official document).

Revenues explicit forecast (no Luminex)



Moleculars revenues (no Luminex)



CLIA market 2020 billion (USD)	7.34
CLIA market 2021 billion (USD)	8.8
CLIA forecast 2028 billion (USD)	13.75
€//\$	1.22
CLIA market 2020 billion (EUR)	6.04
CLIA market 2021 billion (EUR)	7.24
CLIA forecast 2028 billion (EUR)	11.31
CAGR 2021-2028	7.90%

Immunoassay market value 2021 billion (USD)	23.3
Revenue Forecast 2028 billion (USD)	36.7
€//\$	1.22
Immunoassay market vakue 2021 billion (EUR)	19.16
Revenue Forecast 2028 billion (EUR)	30.18
CAGR 2021-2028	6.70%

EIA/ELISA segment out of Immunoassay	0.637
EIA/ELISA segment value	14.8421
elisa growth≠immunoassays growth	

Euro per kit erosion (*)	2013-2017	2017-2022
Clia Δp CAGR	-0.80%	-0.80%
Elisa Δp CAGR	-0.50%	-0.60%
media mercato	-0.50%	-0.60%

DCF Valuation II

The same logic has been applied to Luminex in order to obtain the sales of the company.

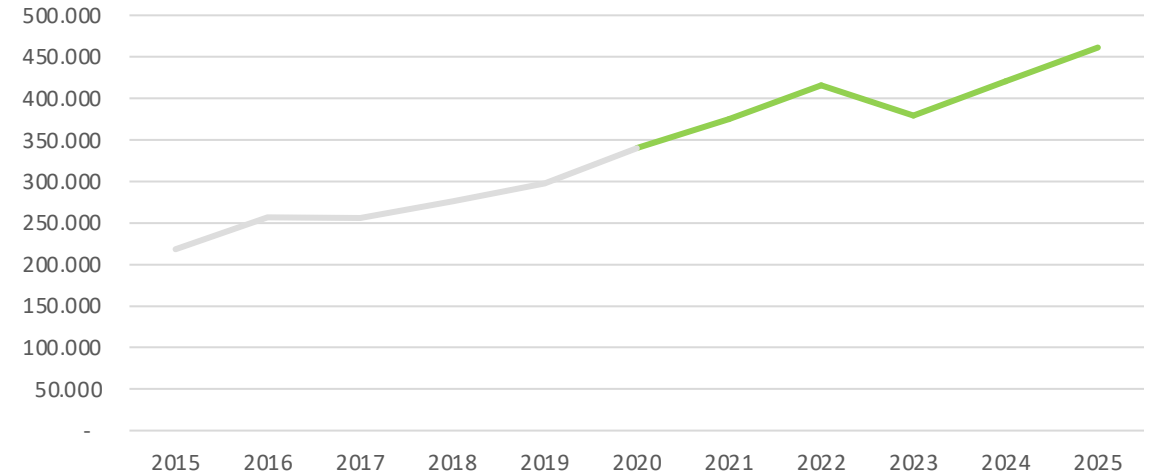
Other Components

Working capital has been computed by dividing between his components and by calculating rotation indexes such as average collection time, average payables time. **No correction to the historical mean** has been applied to average collection time, average payables time and rotation of the inventory since during 2015-2020 **values** were **stable** and insensible to different values of sales **neither** we forecast an increase in **time dilations** or **different strategies for inventory**. In 2030 Amortization/Capex is (83%).

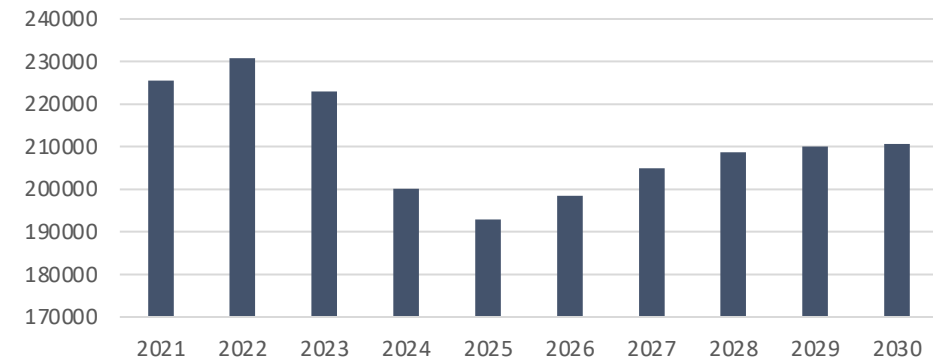
Consideration on continuing value

The explicit forecast used is 2021-2025, the implicit forecast of 10 years contains a reversion to the long term values.
 $CV/EV=72\%$

Revenues Luminex



Unlevered free cash flows



DCF Results

The steady state growth is estimated on the basis of market and macroeconomics consideration. **High growth** in the market has to be expected because of continues growth in the health care sector due to:

1. For developed countries **increasing demand of well being** and **aging** of the population
2. Emerging countries need to catch up in the healthcare sector

From our considerations acquisition of Luminex should have brought an increase in **VPS of 8 €** while the price reacted on the day of acquisition of 13,45 € to later smaller values.

		Steady growth						
WACC	VPS	3,25%	3,50%	3,75%	4,00%	4,25%	4,50%	4,75%
	5,96%	191,33	207,13	226,50	250,82	282,25	324,44	384,06
	6,17%	175,16	188,26	204,07	223,51	248,01	279,83	322,83
	6,39%	161,20	172,20	185,28	201,10	220,63	245,33	277,57
	6,60%	149,04	158,36	169,32	182,38	198,23	217,85	242,77
	6,81%	138,36	146,33	155,61	166,54	179,60	195,48	215,21
	7,03%	128,90	135,77	143,70	152,94	163,84	176,89	192,82
	7,24%	120,46	126,43	133,25	141,13	150,33	161,21	174,27

		Steady state residual earnings growth rate						
WACC	EV/EBITDA	3,25%	3,50%	3,75%	4,00%	4,25%	4,50%	4,75%
	5,96%	22x	24x	26x	28x	32x	36x	43x
	6,17%	20x	22x	24x	26x	28x	32x	36x
	6,39%	19x	20x	22x	23x	25x	28x	31x
	6,60%	18x	19x	20x	21x	23x	25x	28x
	6,81%	17x	17x	18x	19x	21x	23x	25x
	7,03%	16x	16x	17x	18x	19x	21x	22x
	7,24%	15x	15x	16x	17x	18x	19x	20x

Residual Earnings Model - Assumptions

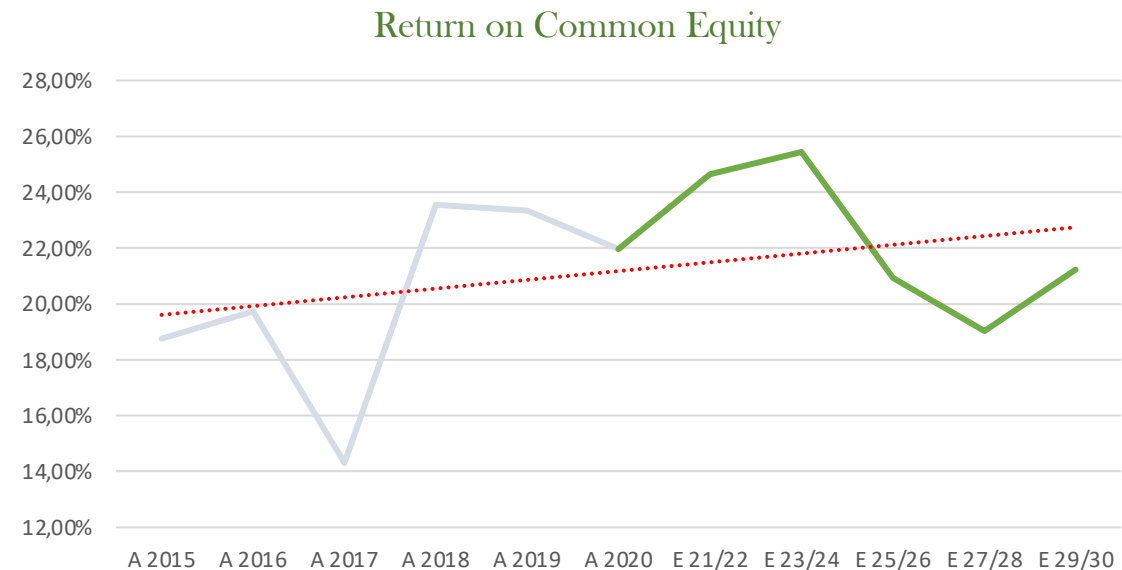
As to estimate the evolution of the profitability through the years (key element in the RE model), we used a **simple moving average** with a peculiarity: **the time serie is reversed** (the first element of the serie is the most recent data, the last element is the least recent one).

The computation of the estimate does not rely on a strict mathematical framework (**backward moving average is not used to forecast usually**), but rather on a realistic estimate of the immediate future: we believe the better way to estimate the profitability in the short run imply relying more on the most recent data, while the medium run will present a **mean reversion** to lower values (as the covid revenues will slowly fade away).

Due to the fact that DiaSorin is an expanding company, we reckon that the mean reversion will not be towards the mean value of profitability of the historical dataset, so we applied a **premium** of 2% on the estimate due to the cost synergy between the two firms and managerial assumptions.

$$E(\text{Profitability}_{2021}) = \text{Mean}(\text{ROCE}_{2019}, \text{ROCE}_{2020}) + \text{premium}$$

$$E(\text{Profitability}_{2030}) = \text{Mean}(\text{ROCE}_{2015}, \text{ROCE}_{2016}) + \text{premium}$$



Residual Earnings Model - Results

To complete the set up of the model a **dividend payout** of 40.67%, fixed all along the forecasted horizon of ten years, a **steady state residual earnings growth** of 4.00% and a **required return** of 7.09% were used.

The 79.88% of the value implied by the model is given by the continuing value due to the high growth rate of the company over the forecasted horizon.

As it can be seen from the sensitivity analysis, a reasonable range of the value per share goes from 151.66 € to 206.68 €

		Steady state residual earnings growth rate						
Required return	VPS	3.25%	3.50%	3.75%	4.00%	4.25%	4.50%	4.75%
	6.34%	193.01	205.82	221.10	239.64	262.62	291.84	330.25
	6.59%	176.26	186.74	199.07	213.78	231.63	253.75	281.88
	6.84%	161.89	170.57	180.66	192.52	206.68	223.86	245.15
	7.09%	149.43	156.70	165.05	174.76	186.17	199.79	216.32
	7.34%	138.53	144.67	151.66	159.70	169.04	180.02	193.12
	7.59%	128.93	134.15	140.05	146.78	154.50	163.48	174.04
	7.84%	120.41	124.88	129.90	135.57	142.04	149.47	158.10

Reverse Engineering

Implied RE growth rate	
Book value	17.09
Market value	191.70
Residual Earnings at t + 1	3.00
Required return	7.09%
Implied growth rate	5.37%

Given the cost of equity we have estimated, through a reverse engineering of a simple valuation model with residual earnings, we inferred that the **market expects** DiaSorin to have a **steady state growth of 5.37%**, so compatible with our computations.

In the figure on the right, we have performed a valuation based on book value and near-term forecasts, without any growth in the long term.

The differential between the value obtained and the current market value is the **value of speculative growth financial markets are available to pay**, currently **62.81%** of the market value of the firm.

	2020 A	2021 E	2022 E	2023 E	2024 E	2025 E
EPS	3.54	4.21	4.83	5.71	6.58	6.23
DPS	0.93	1.71	1.96	2.32	2.67	2.53
BPS	17.09	19.59	22.46	25.85	29.75	33.45
Required return		7.09%	7.09%	7.09%	7.09%	7.09%
Residual Earnings (RE)		3.00	3.44	4.12	4.74	4.12
PV of RE		2.80	3.00	3.36	3.61	2.92
Total PV of forecasted RE	15.69					
Continuing value						54.23
Present value of CV	38.50					
Share value	71.29	Share value in the absence of growth				
Current market price	191.70					
Value of speculative growth (in €)	120.41					
Value of speculative growth (in %)	62.81%	Part of the market value based on speculative growth				

Peers (in vitro diagnostic)	Business Activity
Exact Sciences Corp	Cancer detection
BioMerieux	Diagnostic (COVID), industrial microbiology for food, cosmetic and pharmaceutical industry
QIAGEN NV	Diagnostic (COVID), molecular testing (start to finish)
Myriad Genetics Inc	Genetic Diagnostic
Meridian Bioscience Inc	88% Diluted EPS Growth (different stage)
OraSure Technologies Inc	Diagnostic (COVID), drug testing, different stage
Ortho Clinical Diagnostic	Blood: transfusion, clinical chem, consulting
Tecan Group AG	Different diagnostics, many products and automatization solutions

During the research of the peers, we met some difficulties:

- Crean (2016) states that «given the complexities and uncertainties [...], it is rare to find two companies or assets sufficiently alike to be directly comparable»
- Most competitors are conglomerates, where the similar business lines are just a small part of the company (AbbVie, Roche, etc.)
- Peers with the same size in the in-vitro diagnostic market often specialize (e.g., only cancer detection or blood testing)

Therefore, we decided to include companies in the in-vitro diagnostic sector with similar size to get a broad overview of the range of values implied by the sector.

Due to the described factors, it should be seen as a first overview and not as standalone valuation

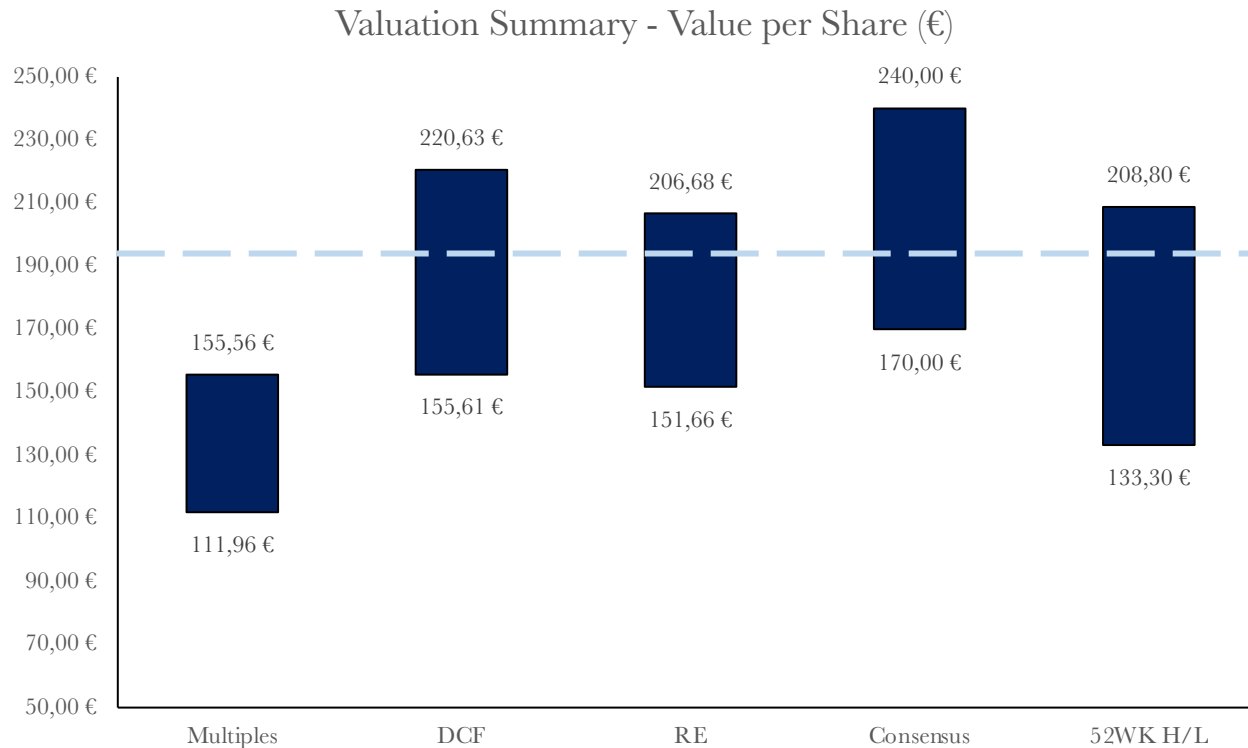
Multiples Valuation

Peers (in vitro diagnostic)	Market Cap (bn)	EV (bn)	Currency	P/E	P/S	EV/ADJ EBITDA	P/LFCF
Exact Sciences Corp	17.125	18.29	USD	--	9.5	--	--
BioMerieux	12.469	12.552	EUR	24.5	3.9	13.5	55.8
QIAGEN NV	12.072	13.055	USD	20.8	5.5	16.5	29
Myriad Genetics Inc	2.525	2.541	USD	40.5	4.2	--	--
Meridian Bioscience Inc	0.781	0.767	USD	11.1	2.6	7.0	20.25
OraSure Technologies Inc	0.696	0.616	USD	--	2.9	32.0	--
Ortho Clinical Diagnostic	4.752	6.76	USD	--	2.4	30.3	11.78
Tecan Group AG	6.856	6.881	CHF	43.5	7.4	32.5	41.03
Mean value				28.1	4.8	22.0	31.6
DiaSorin implied share price / EV (EUR)				155.56	111.96	9.608 bn	122.82
DiaSorin actual share price / EV (EUR)				191.70		9.793 bn	

The data on which the calculations are based on is mainly retrieved from Bloomberg. TTM values were used.

- This valuation implies share values from 111.96€ up to 155.56€ compared to an actual value of 191.70€
- A COVID revenue premium might have been priced in DiaSorin market value, while the average sector peer does not have it
- In addition, the average sector peer did not have a huge acquisition mid 2021

Football Field and Recommendation



As it is depicted by the football field, excluding the multiples valuation, that is inappropriate due to the lack of truly peers, we recommend a **HOLD**, as it cannot be inferred a clear signal of the stock mispricing. Different conclusions can be done based on its negative correlation to the covid pandemic: an investor exposed to it could diversify with this its stock.



HOLD

Thank you

Any Questions?